A Project Report On

"Workout Meter"

For the Partial Fulfilment of the Award of the Degree of Bachelor of Computer Application of Veer Narmad South Gujarat University, Surat.

Bachelor of Computer Application [B.C.A]
Semester – VI

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Certificate

This is to certify that **Shaw Aish B.** Exam Seat Number: **3805**

have / has satisfactorily presented their Project work Entitled **"*Workout Meter*"** as a partial fulfillment of requirements for **6th *Semester* – *B.C.A.,*** during the academic Year 2026-2027.

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Thank you

Shaw Aish B.

Singh Shivam N.

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CHAPTER 1: INTRODUCTION

In the modern digital era, maintaining a healthy lifestyle has become a major priority for individuals across all age groups. With increasing awareness about physical fitness, people are actively engaging in gym activities, workouts, and health routines. However, maintaining consistency and tracking progress effectively remains a challenge for many individuals. Most people either rely on memory or use traditional methods such as notebooks to record their daily workouts, which is neither efficient nor reliable.

The need for a proper system that can manage and track workout data in an organized manner has led to the development of digital solutions. The **Workout Meter** project is designed as a web-based application that provides users with an efficient and centralized platform to manage their fitness activities. This system is developed using the **MEAN Stack**, which includes MongoDB, Express.js, Angular, and Node.js.

The application is designed with a **multi-dashboard system**, consisting of three main dashboards: Admin, User, and Seller. Each dashboard is responsible for performing specific tasks. The Admin dashboard controls the entire system, the User dashboard allows individuals to track workouts and purchase gym supplements, and the Seller dashboard helps manage products and orders.

Another important feature of this system is its integration of e-commerce functionality. Users can browse, select, and purchase gym supplements directly through the platform. This eliminates the need for separate applications and provides a complete fitness solution in one place.

The system also ensures data security and accessibility. All data is stored in a centralized database, allowing users to access their information anytime and from any device. The use of modern technologies ensures that the system is fast, scalable, and user-friendly.

In conclusion, Workout Meter is a comprehensive solution that combines fitness tracking with product management, making it a practical and efficient application for modern users.

1.1 Existing System

The existing system for workout tracking is mostly manual or limited to basic applications. Many users still prefer maintaining physical records such as notebooks or diaries to track their daily workouts. While this method is simple, it lacks efficiency and is not suitable for long-term tracking.

Some digital solutions are available in the form of mobile applications. However, these applications often provide only basic features such as logging workouts without offering advanced functionalities like analytics, dashboards, or integration with product purchasing systems.

Another major issue with existing systems is the lack of a centralized platform. Users have to rely on multiple applications or methods to manage different aspects of fitness, such as workout tracking, diet management, and supplement purchasing. This leads to inconvenience and inefficiency.

1.2 Limitation of Existing System

The existing system suffers from several significant limitations:

- **Manual Effort:** Users need to manually enter and maintain records, which is time-consuming.
- **Data Inaccuracy:** High chances of errors due to manual entry.
- **Lack of Security:** Data stored in notebooks or unsecured apps can be lost or misused.
- **No Real-Time Tracking:** Users cannot monitor progress instantly.
- **No Integration:** Workout tracking and product purchasing are handled separately.
- **Poor User Experience:** Most systems lack interactive dashboards and modern UI.

These limitations highlight the need for a more advanced and integrated system.

1.3 Proposed System Introduction

The proposed system, Workout Meter, is designed to overcome the limitations of the existing system by providing a complete and integrated solution. It offers a centralized platform where users can track their workouts, monitor progress, and purchase gym supplements.

Workout Meter, is a web-based application developed using the MEAN stack. It provides a centralized and efficient platform for managing workout activities and fitness-related data.

The system is designed with a **multi-dashboard architecture**, which includes:

- Admin Dashboard
- User Dashboard
- Seller Dashboard

Each dashboard has specific roles and responsibilities. The admin manages users and sellers, the user tracks workouts and purchases supplements, and the seller manages products and orders.

This system ensures better performance, improved user experience, and efficient data management.

1.4 PROJECT PROFILE

Workout Meter is a full-stack web application that includes three major modules:

1. Admin Module

- Manages users and sellers
- Monitors system activities
- Can block/unblock users or sellers

2. User Module

- Tracks workouts
- Adds new workouts with gym details
- Purchases gym supplements
- Views cart and order history

3. Seller Module

- Manages products
- Handles customer orders
- Tracks buyer details

This modular structure ensures better organization and scalability of the system.

1.5 Scope of Proposed System

The scope of the system includes:

- Providing a platform for tracking workouts
- Enabling users to purchase gym supplements
- Allowing sellers to manage products and orders
- Providing admin control for monitoring activities

The system can be expanded to include additional features such as AI-based fitness recommendations and mobile applications.

1.6 Objective of Proposed System

The objectives are:

- To provide an efficient workout tracking system
- To ensure data security and reliability
- To enhance user experience
- To integrate fitness tracking with e-commerce

1.7 System Environment Description

- Frontend: Angular 17
- Backend: Node.js with Express.js
- Database: MongoDB

CHAPTER 2: TECHNOLOGY USED

2.1 What is Angular 17

Angular 17 is a modern frontend framework developed by Google for building dynamic web applications. It follows a component-based architecture, where the application is divided into smaller reusable components.

Angular provides features such as two-way data binding, dependency injection, and modular structure. These features help in building scalable and maintainable applications.

In Workout Meter, Angular is used to design the user interface and manage user interactions efficiently.

2.2 Node.js and Express.js

Node.js is a runtime environment that allows JavaScript to be used for server-side development. It is known for its speed and efficiency, especially in handling multiple requests simultaneously.

Express.js is a framework built on top of Node.js that simplifies the process of building web applications. It provides features such as routing, middleware, and API handling, making backend development easier and faster.

The server plays a crucial role in handling requests from the client side. It processes data, performs business logic, and communicates with the database.

Node.js is used as the server in this project because it is fast and efficient. Express.js is used along with Node.js to simplify API development and routing.

The server ensures that all operations such as login, data storage, and order processing are handled securely and efficiently.

2.3 MongoDB

MongoDB is a NoSQL database that stores data in a flexible, JSON-like format. Unlike traditional databases, it does not require a fixed schema, allowing developers to modify data structures easily.

It is highly scalable and suitable for applications that handle large amounts of data.

In this project, MongoDB is used to store user data, workout details, product information, and order records.

2.4 HTML

HTML is used to create the structure of web pages. It defines elements such as headings, paragraphs, forms, and buttons.

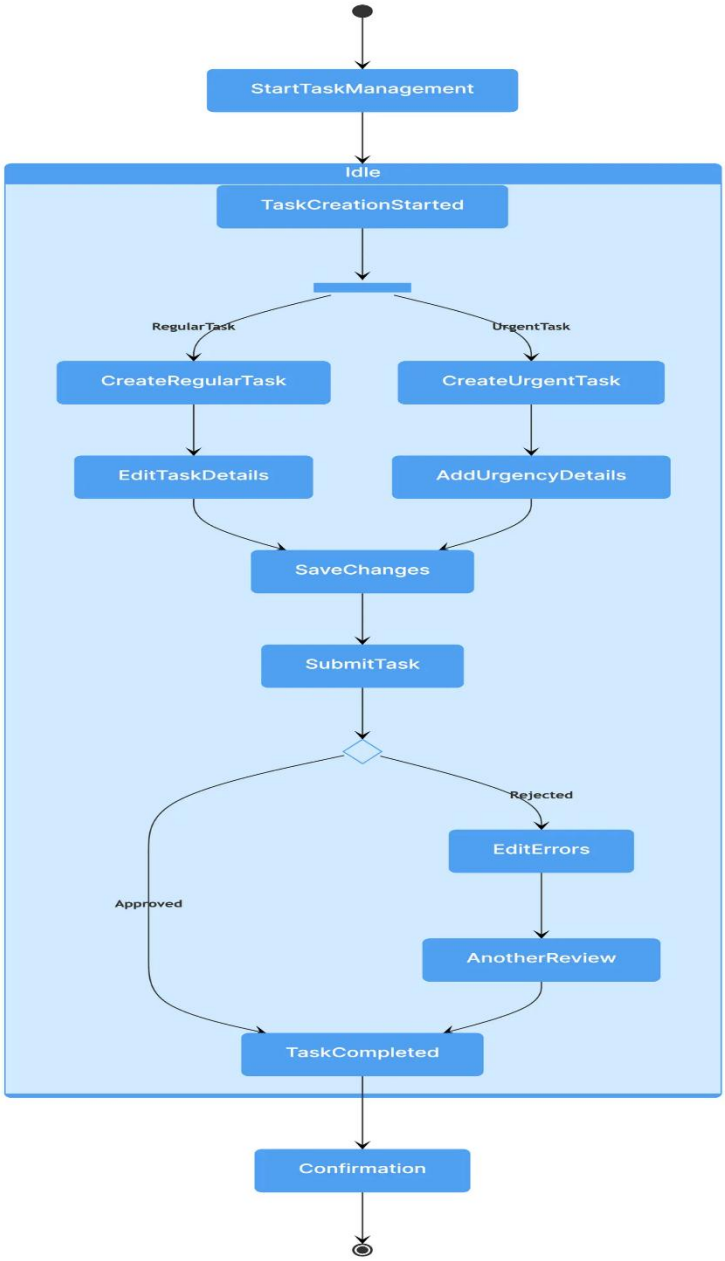
2.5 CSS, Bootstrap, and Tailwind

These technologies are used to design the user interface:

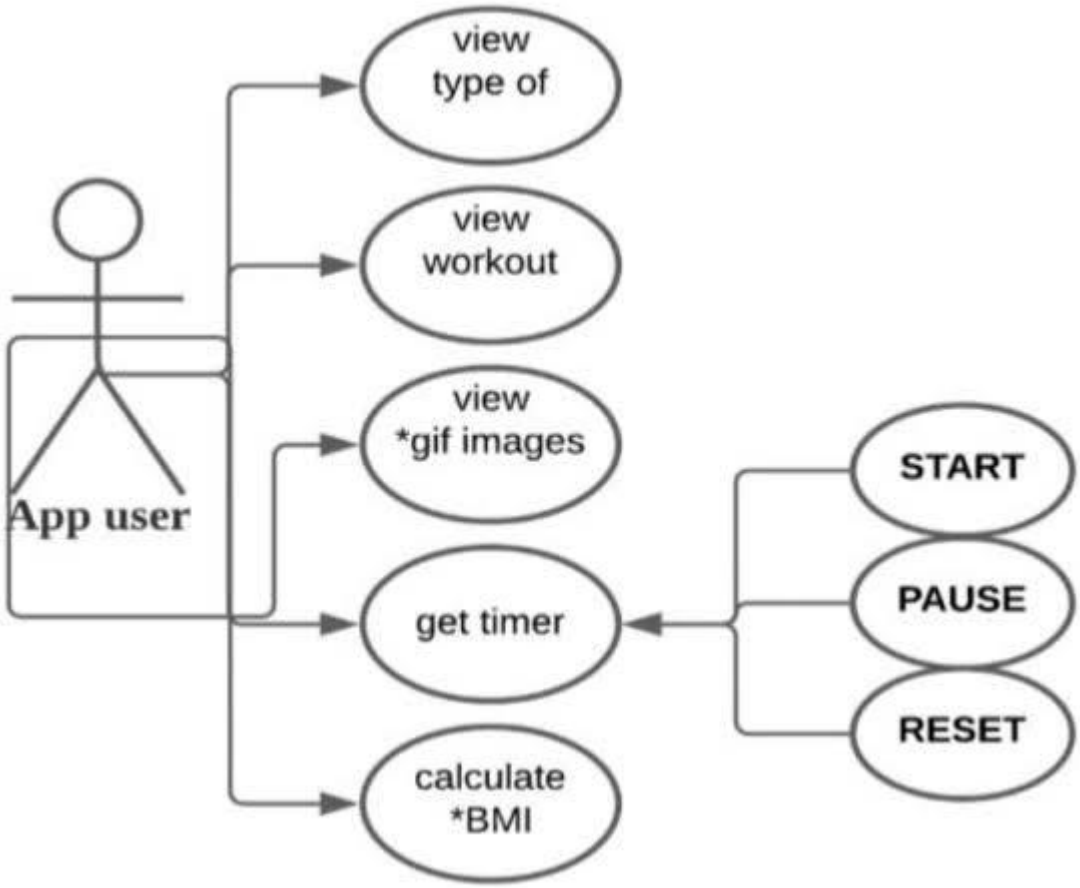
- CSS provides styling
- Bootstrap offers ready-made components
- Tailwind allows utility-based design

CHAPTER 3: SYSYTEM DESIGN

3.1 Task Dependency Diagram



3.2 Data Flow Diagram (DFD)/UML



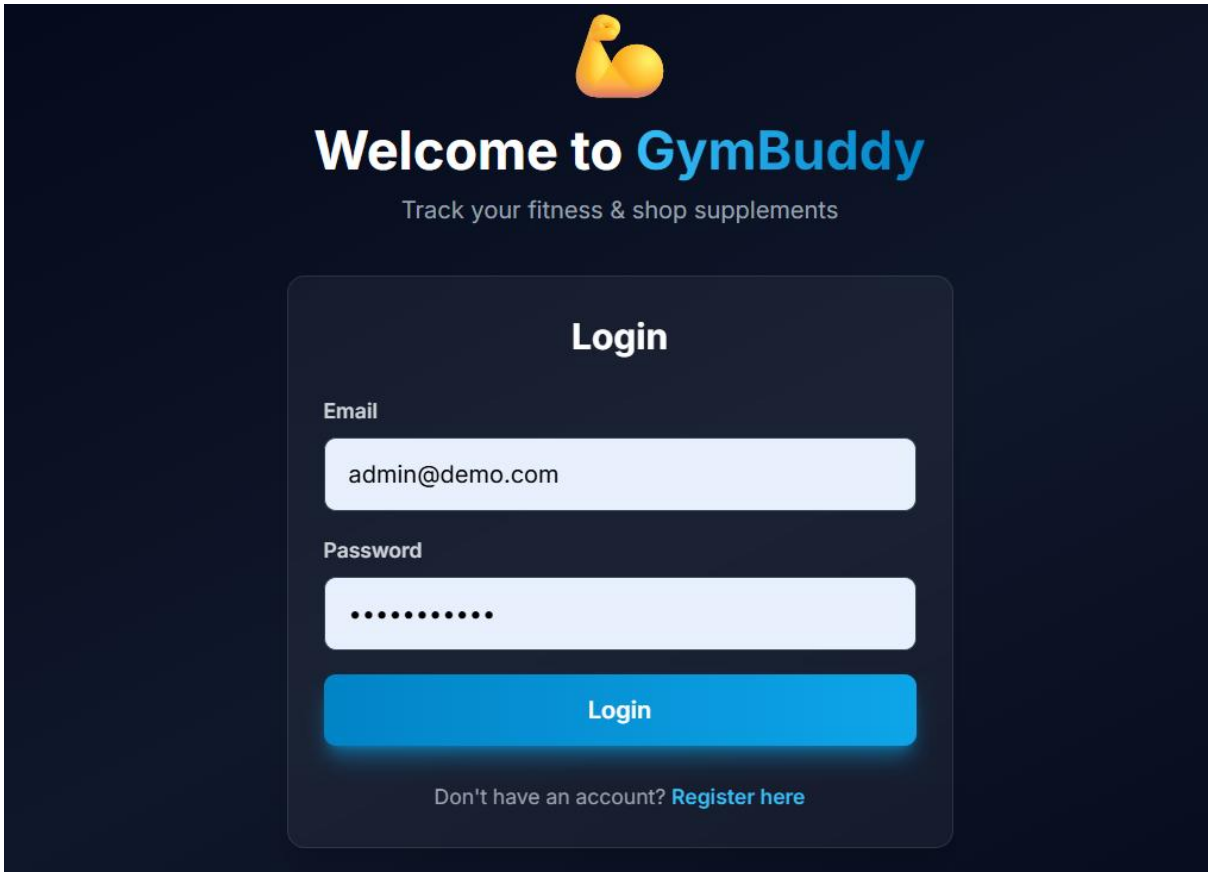
3.3 Data Tables

Table	Fields
User	ID, Name, Email
Seller	ID, Products
Product	ID, Price
Order	ID, UserID

CHAPTER 4: USER INTERFACE

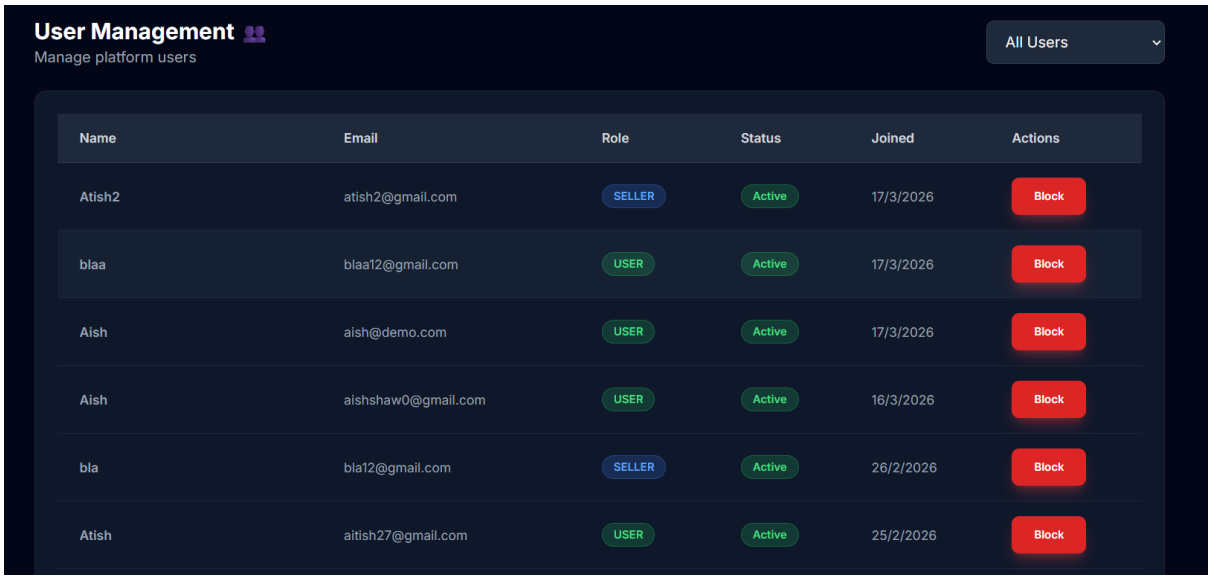
4.1 Admin Side

4.1.1 Admin Login



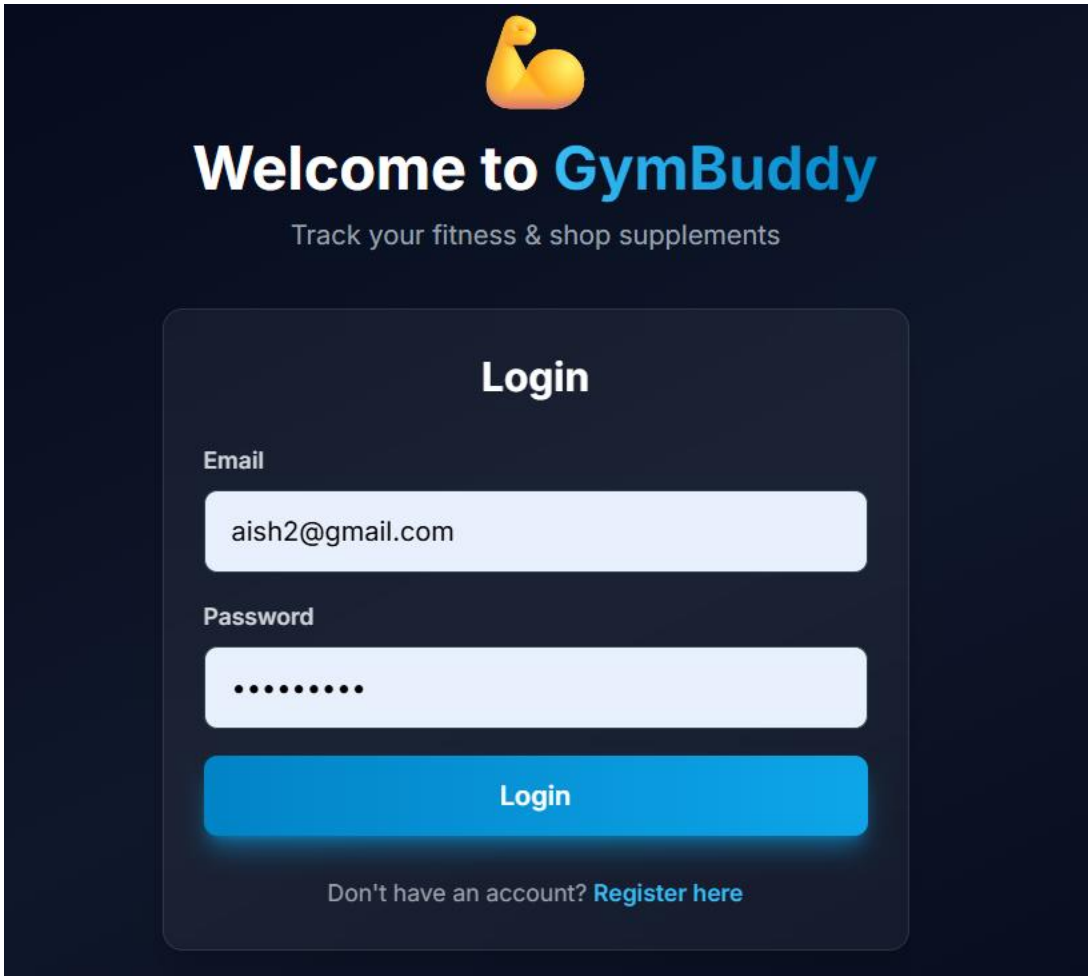
4.1.2 Admin Dashboard

4.1.3 User & Seller Management

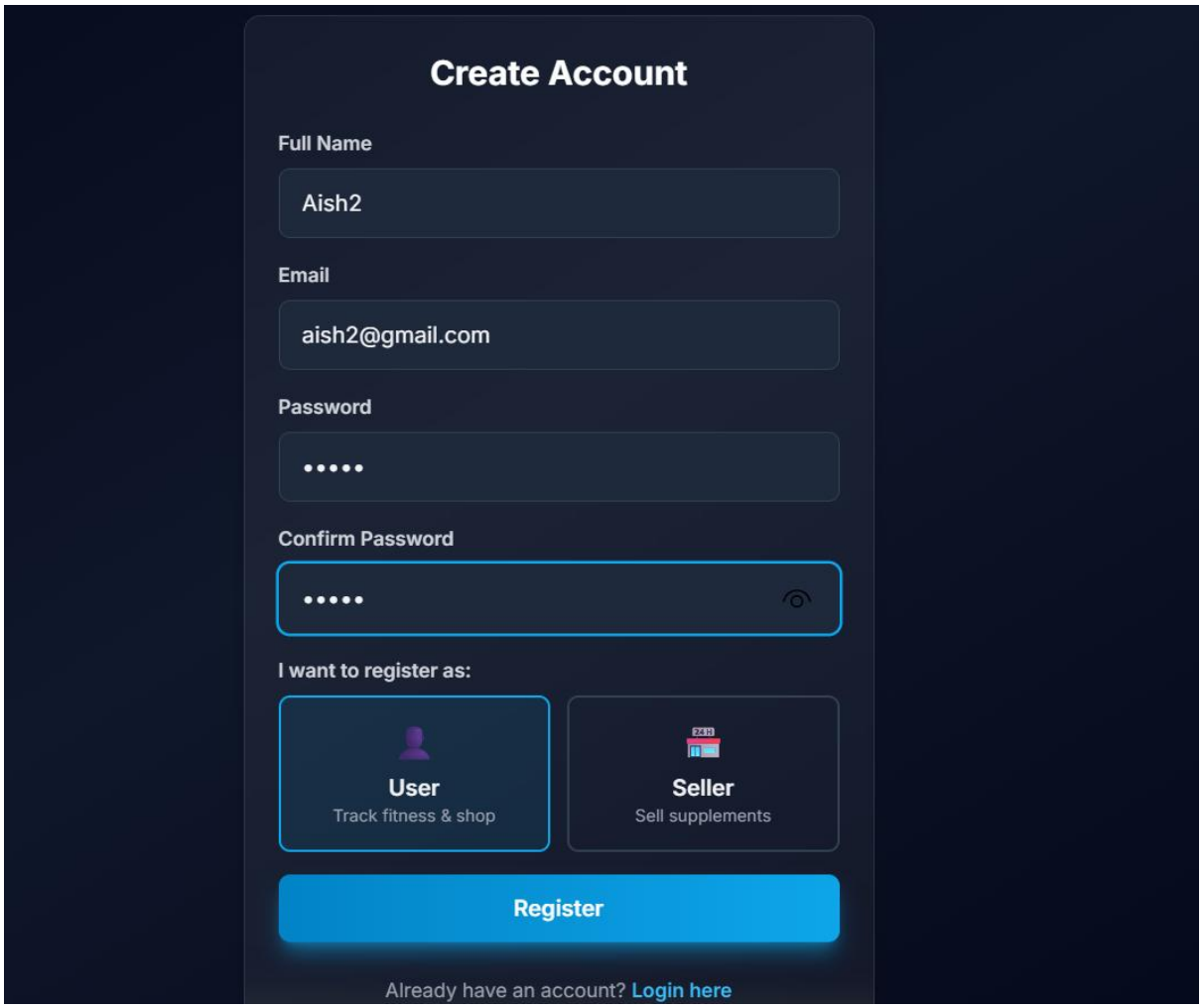


4.2 Client Side

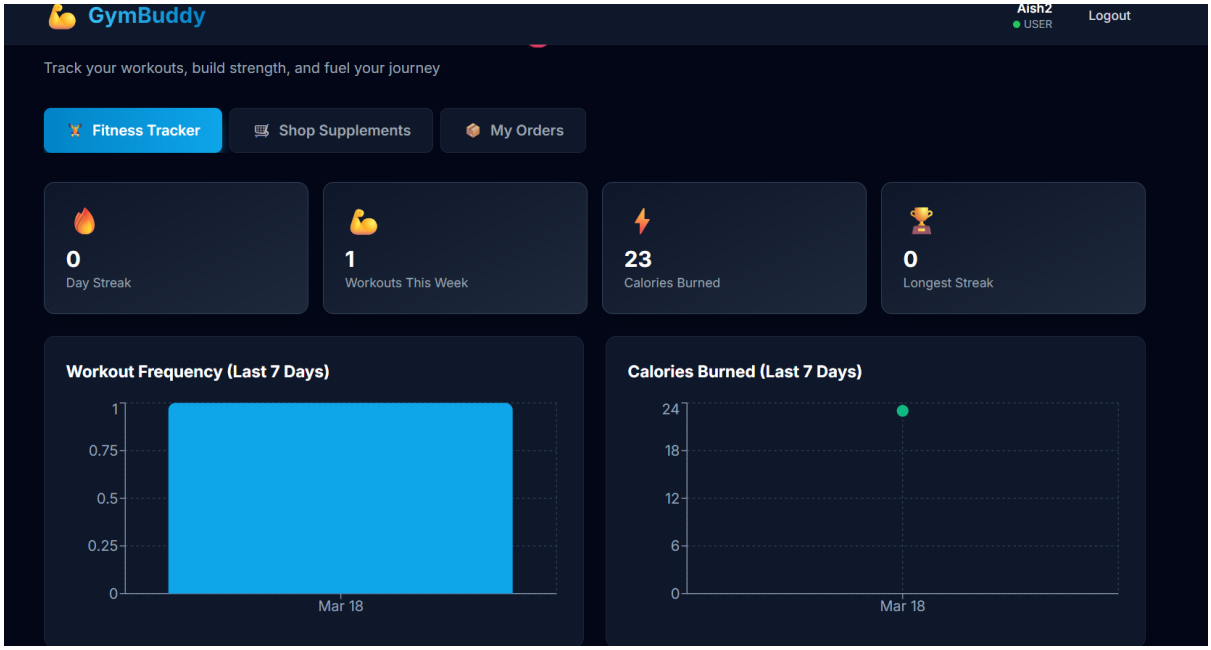
4.2.1 Client Login



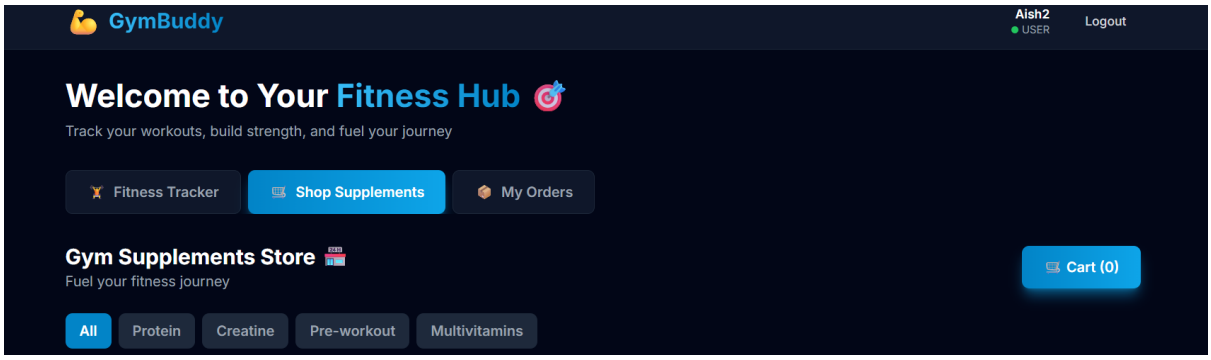
4.2.2 Client Registration

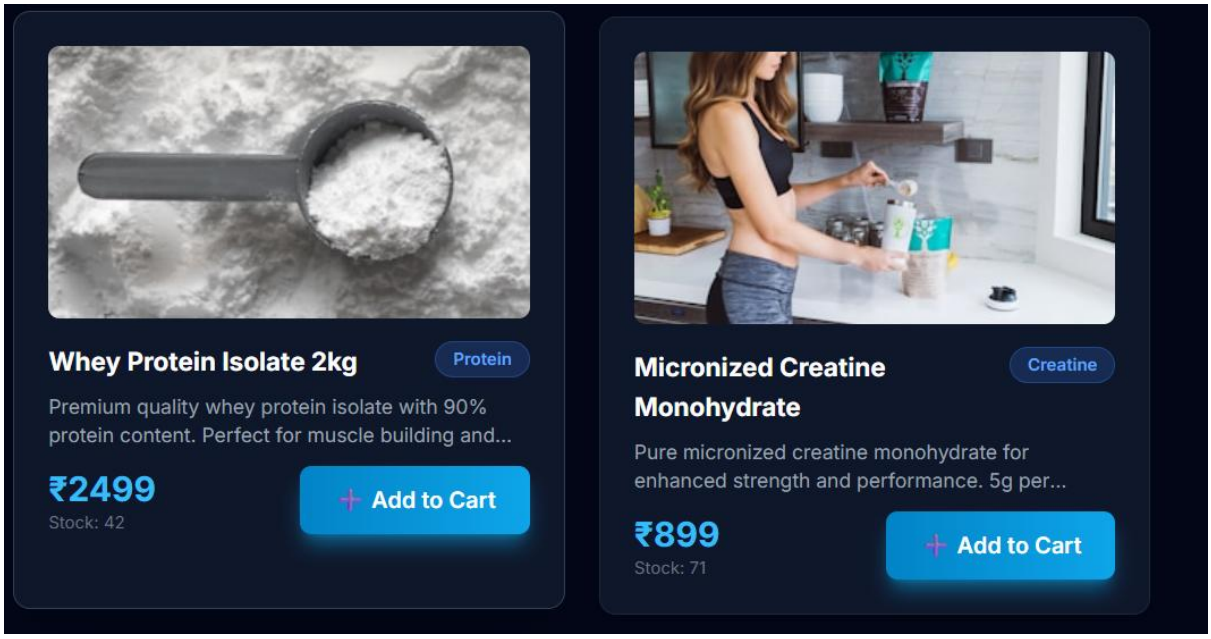


4.2.3 Client Dashboard



4.2.4 Product Store

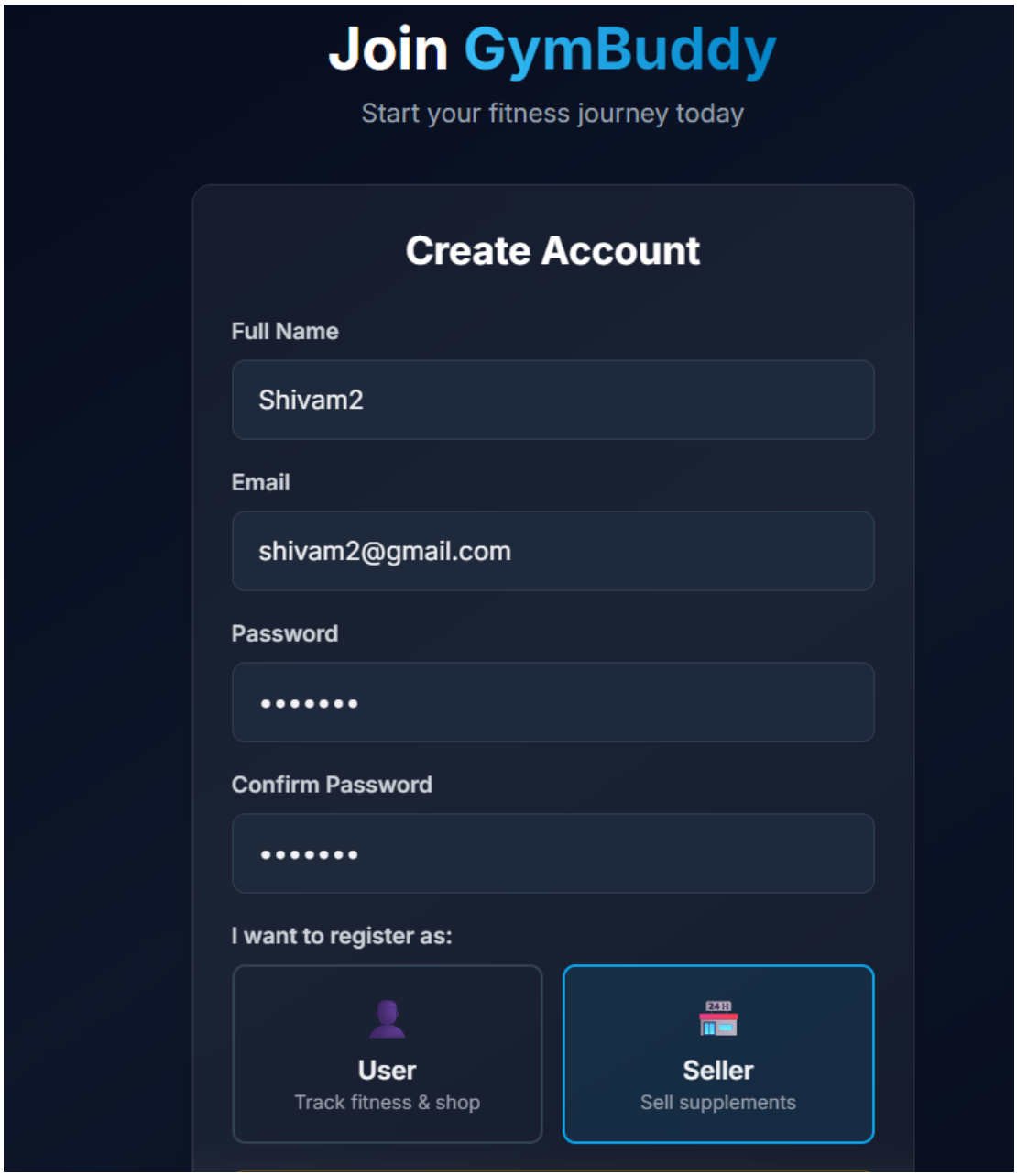




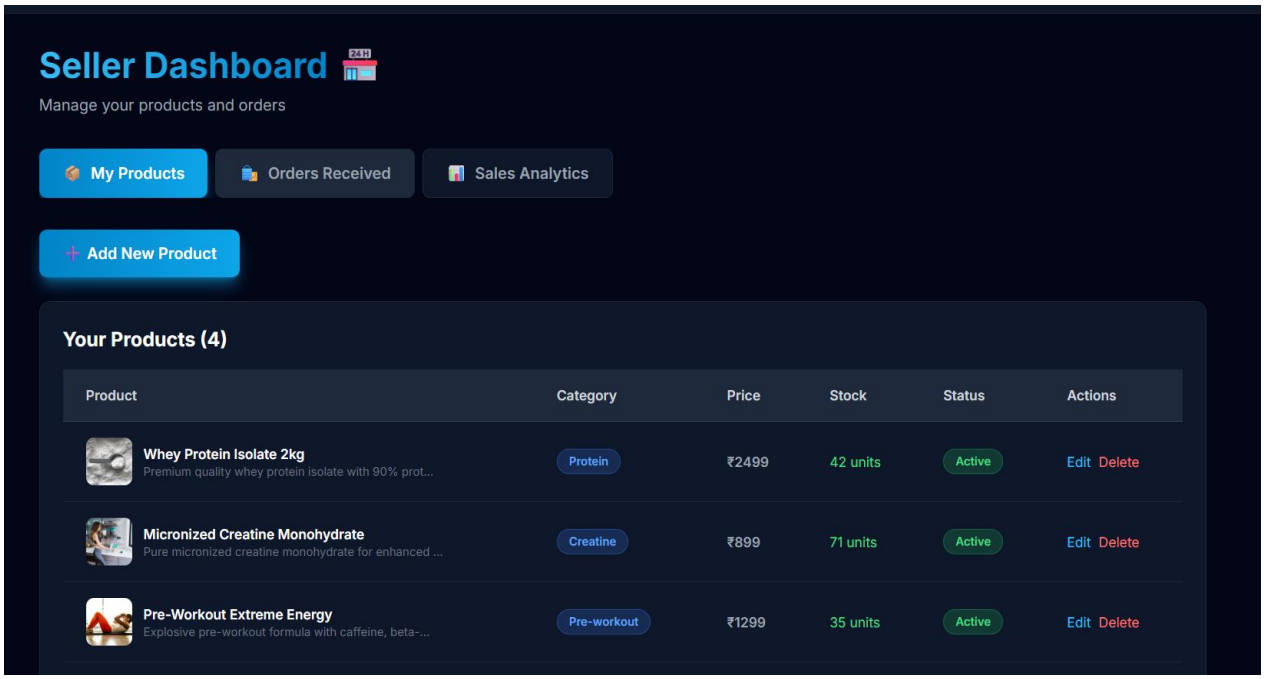
4.3 Seller Side

4.3.1 Seller Login

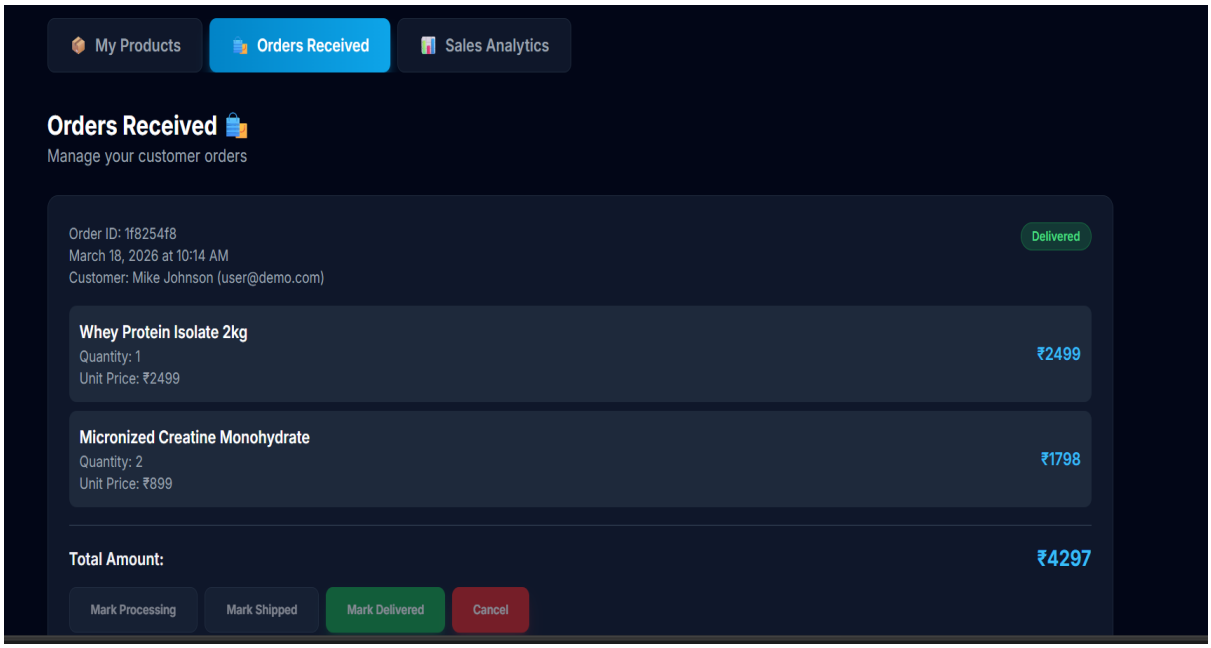
4.3.2 Seller Registration



4.3.3 Seller Dashboard



4.3.4 Order Management



CHAPTER 5: CONCLUSION

The Workout Meter project successfully demonstrates the development of a modern web-based application using the MEAN stack technology. The system has been designed to provide a complete solution for managing fitness activities along with an integrated platform for purchasing gym supplements. By combining multiple functionalities into a single application, the project addresses the limitations of traditional workout tracking methods and offers a more efficient and user-friendly alternative.

One of the major strengths of this project is its multi-dashboard architecture, which includes Admin, User, and Seller modules. Each module is designed to perform specific tasks, ensuring better organization, improved system performance, and enhanced security. The Admin dashboard provides complete control over users and sellers, allowing monitoring and management of the entire system. The User dashboard enables individuals to track their workouts, maintain records, and purchase supplements, while the Seller dashboard allows efficient management of products, orders, and buyer details.

The system also ensures proper data management by using MongoDB as the database, which allows flexible and scalable storage of information. The use of Angular for frontend and Node.js with Express.js for backend ensures smooth communication between different components of the application. This results in a responsive, fast, and reliable system.

In addition to functionality, the project focuses on user experience by providing a clean and interactive interface. The integration of workout tracking with e-commerce features makes the application more practical and useful in real-life scenarios. Users can not only manage their fitness routines but also purchase related products within the same platform.

However, like any system, there is scope for future improvements. Features such as mobile application development, advanced analytics, AI-based workout suggestions, and integration with wearable devices can further enhance the system.

In conclusion, the Workout Meter project achieves its primary objective of creating a comprehensive fitness management system. It provides an effective, scalable, and user-friendly solution that can be further expanded to meet future technological advancements and user needs.

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- **MDN Web Docs (HTML, CSS, JavaScript)**
  https://developer.mozilla.org/

THANK YOU